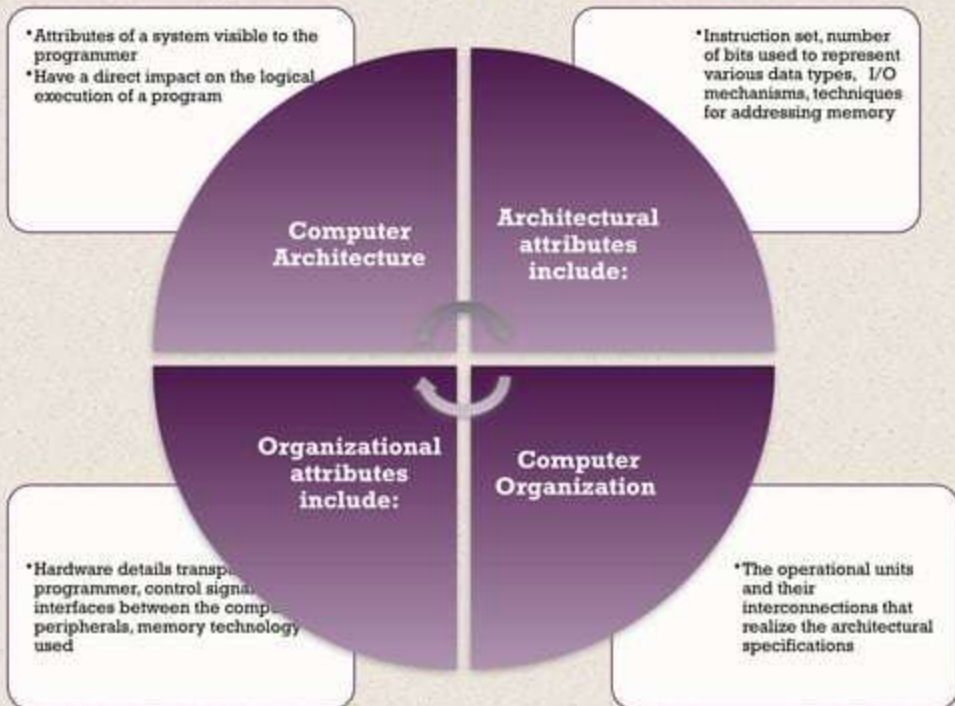


Computer Architecture

Computer Organization



+ Structure and Function



- Hierarchical system
 - Set of interrelated subsystems
- Hierarchical nature of complex systems is essential to both their design and their description
- Designer need only deal with a particular level of the system at a time
 - Concerned with structure and function at each level
- Structure
 - The way in which components relate to each other
- Function
 - The operation of individual components as part of the structure





Function

A computer can perform four basic functions:

- Data processing
- Data storage
- Data movement
- Control

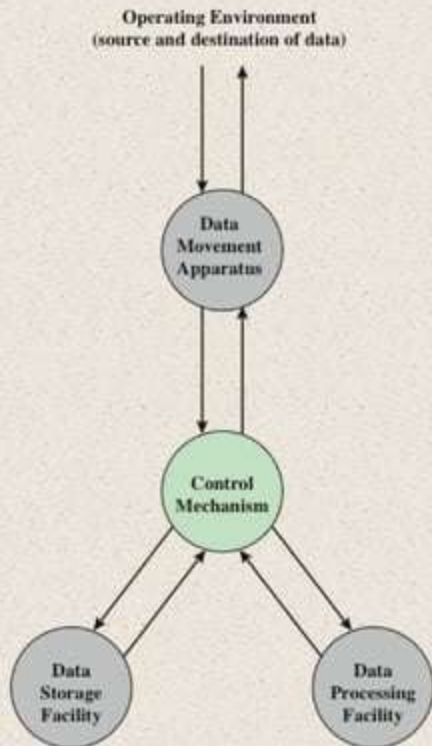
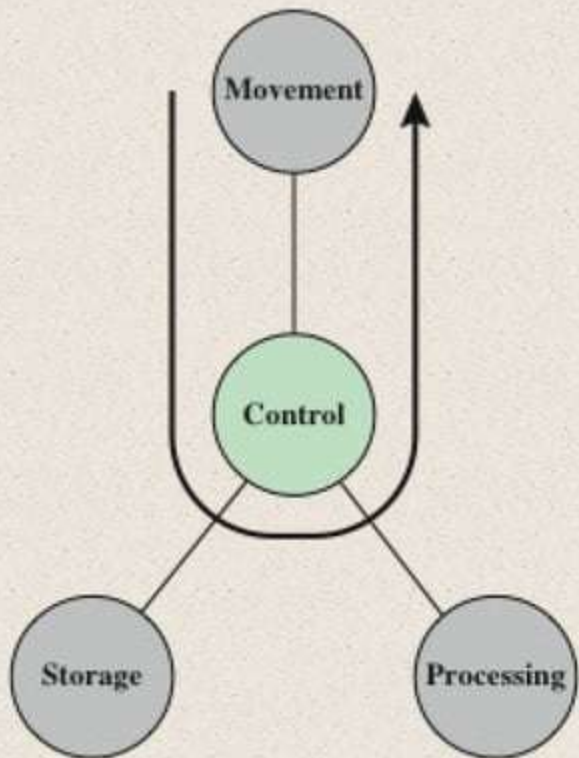


Figure 1.1 A Functional View of the Computer



Operations

(a)
Data movement



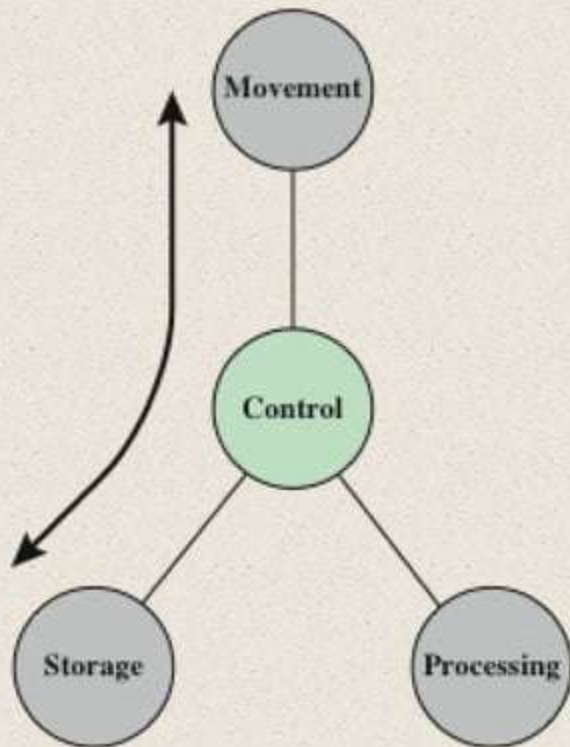
(a)

Figure 1.2 Possible Computer Operations



Operations

(b)
Data storage



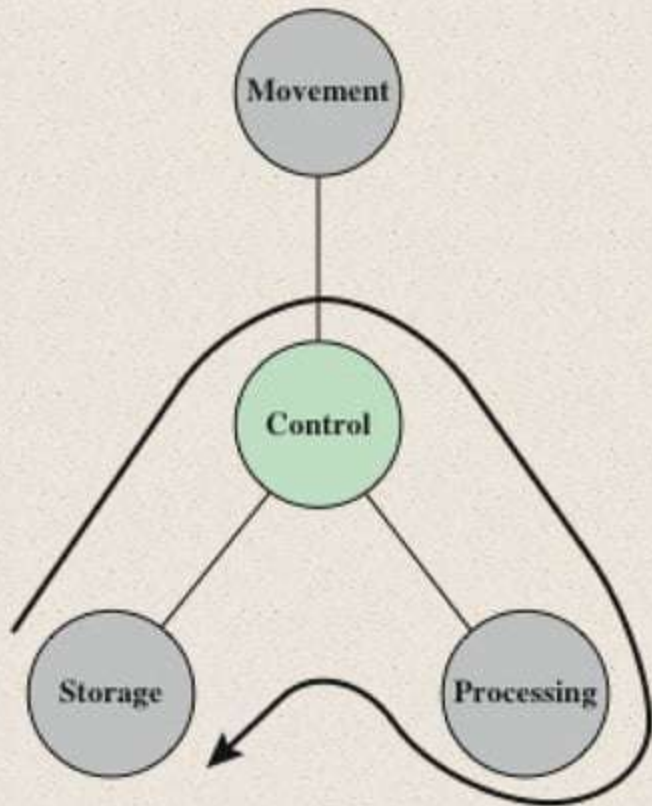
(b)

Figure 1.2 Possible Computer Operations



Operations

(c)
Data processing



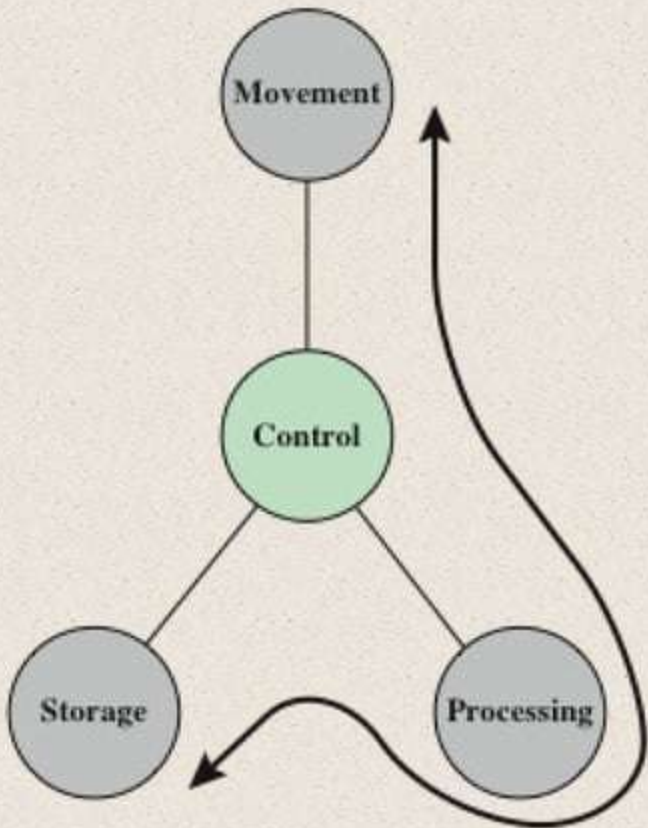
(c)

Figure 1.2 Possible Computer Operations



Operations

(d)
Control



(d)

Figure 1.2 Possible Computer Operations

Structure

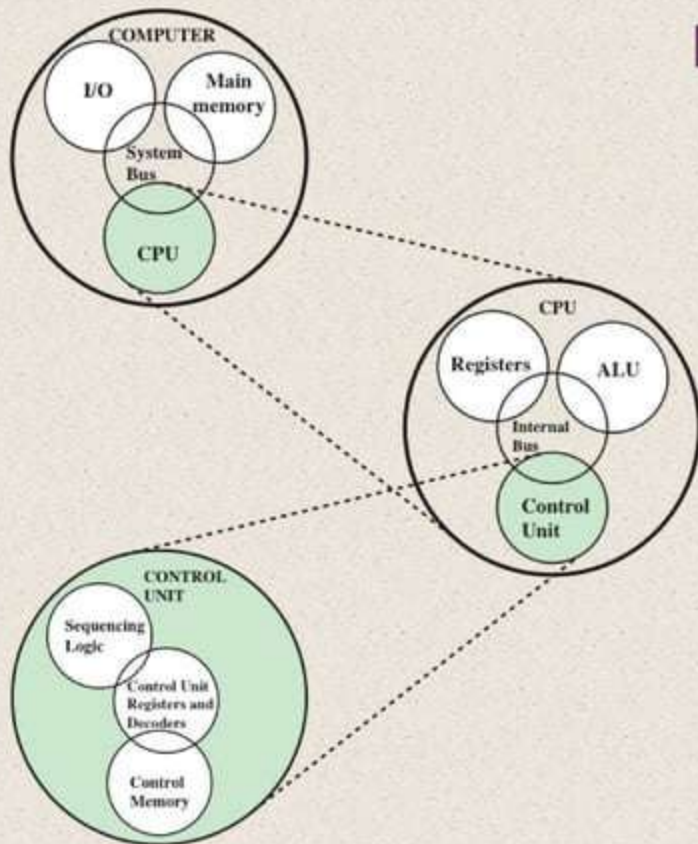


Figure 1.4 A Top-Down View of a Computer



There are four
main structural
components
of the computer:

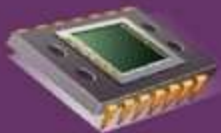


- ✦ CPU – controls the operation of the computer and performs its data processing functions
- ✦ Main Memory – stores data
- ✦ I/O – moves data between the computer and its external environment
- ✦ System Interconnection – some mechanism that provides for communication among CPU, main memory, and I/O



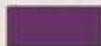
CPU

Major structural components:



- Control Unit
 - Controls the operation of the CPU and hence the computer
- Arithmetic and Logic Unit (ALU)
 - Performs the computer's data processing function
- Registers
 - Provide storage internal to the CPU
- CPU Interconnection
 - Some mechanism that provides for communication among the control unit, ALU, and registers

ENIAC



Weighed
30
tons

Occupied
1500
square
feet
of
floor
space

Contained
more
than
18,000
vacuum
tubes

140 kW
Power
consumption

Capable
of
5000
additions
per
second

Decimal
rather
than
binary
machine

Memory
consisted
of 20
accumulators,
each
capable
of
holding
a
10 digit
number

Major
drawback
was the need
for manual
programming
by setting
switches
and
plugging/
unplugging
cables



John von Neumann

EDVAC (Electronic Discrete Variable Computer)



- First publication of the idea was in 1945
- Stored program concept
 - Attributed to ENIAC designers, most notably the mathematician John von Neumann
 - Program represented in a form suitable for storing in memory alongside the data
- IAS computer
 - Princeton Institute for Advanced Studies
 - Prototype of all subsequent general-purpose computers
 - Completed in 1952

Structure of von Neumann Machine

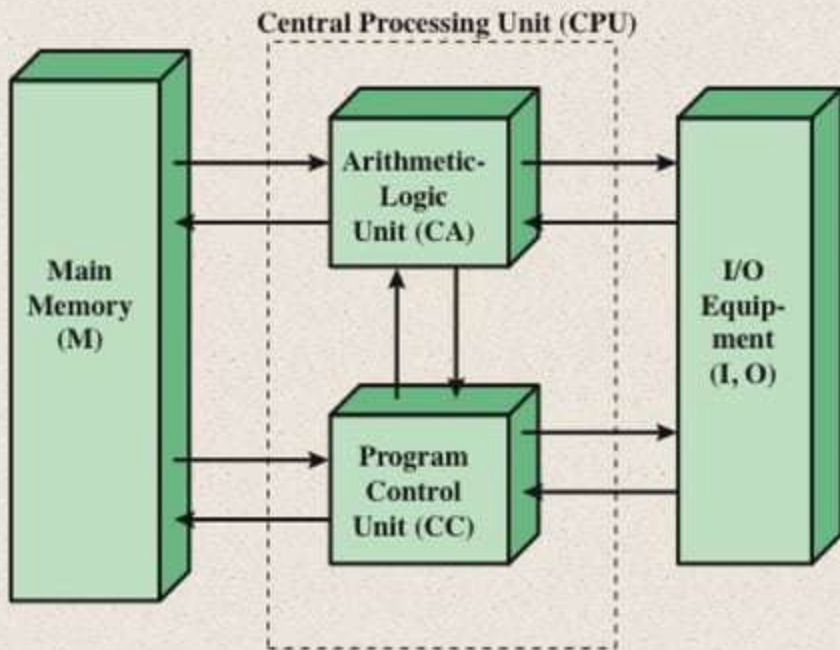


Figure 2.1 Structure of the IAS Computer

+ IAS Memory Formats

- The memory of the IAS consists of 1000 storage locations (called **words**) of 40 bits each
- Both data and instructions are stored there
- Numbers are represented in binary form and each instruction is a binary code

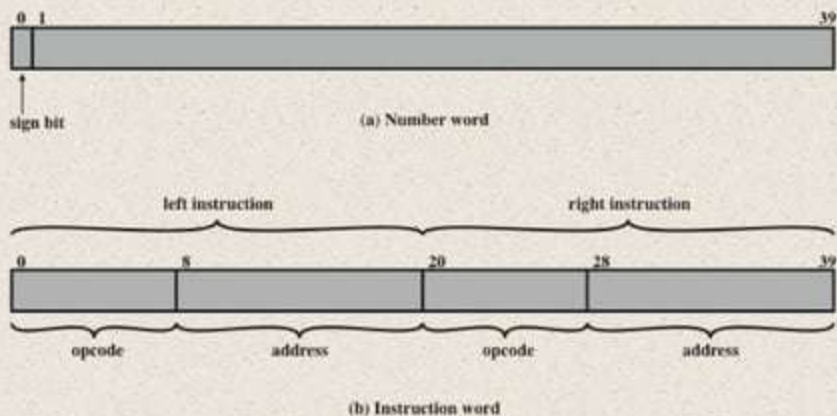


Figure 2.2 IAS Memory Formats



Structure of IAS Computer

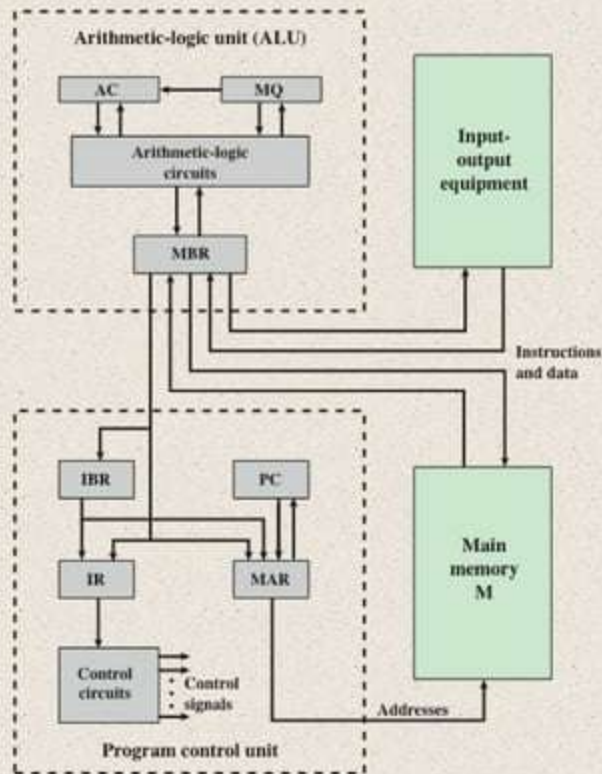


Figure 2.3 Expanded Structure of IAS Computer



Registers

Memory buffer register (MBR)

- Contains a word to be stored in memory or sent to the I/O unit
- Or is used to receive a word from memory or from the I/O unit

Memory address register (MAR)

- Specifies the address in memory of the word to be written from or read into the MBR

Instruction register (IR)

- Contains the 8-bit opcode instruction being executed

Instruction buffer register (IBR)

- Employed to temporarily hold the right-hand instruction from a word in memory

Program counter (PC)

- Contains the address of the next instruction pair to be fetched from memory

Accumulator (AC) and multiplier quotient (MQ)

- Employed to temporarily hold operands and results of ALU operations